Let's understand it using embeddings.

↳ Words are similar if they appear in the same context many times. So, 2 words are likely to be replaceable if they are similar.

honey ← it → bean

"it" has some amount of gravitational pull to both "bean" & "honey".

Once we know the next word "hungry" then both "hungry" & "bean" pulls "it", so "it" gravitate to "bean". & if next word is "delicious" then "it" is pulled by "honey" & "delicious". So, here, "it" describes "honey".

In this way attention works.

↳ A series of transformer blocks is a big NN which predicts the next word, in form of +ve & -ve numbers (scores), so, we need softmax layer.

[since scores can be -ve but prob. can't be negative, so, we do e^{score} & then normalize it (divide by their sum)]

[many feed forward NN with attention layer]

Fine-tuning

The internet is

So, model may ne

Eg. What is t

The m

inter

① Quiz

What is the c of Nigeria?

What is the c of Chad?

What is the of Lebanon

= Next wo

↳ So, model is should be. Re to answer question

So, how do we

The next word

↳ So, we have datasets,

Fine-tuning

The internet is not a question/answer repository.
So, model may not work if it is trained on internet.

Eg. What is the capital of Nigeria? :

The next word may not be "Abuja" on internet. It can be.

① Quiz
What is the capital of Nigeria?
What is the capital of Chad?
What is the capital of Lebanon?

② Story
What is the capital of Nigeria? She asked

③ Chat
What is the capital of Nigeria?
That is the good question.

④ History
What is the capital of Nigeria?
Since 1991 It's Abuja.

(▭ = Next word)

↳ So, model is not very clear, what the next word should be. Remember, transformer model is not trying to answer questions, it's trying to find the next word.
So, how do we get the model to know that it's not just the next word that we want, it's the answer.
↳ So, we have to do is to train it on question-answer datasets.

So, we can use experts to curate a bunch of datasets that have particular questions & answers so that the model starts learning that when it gets a question, it has to answer with the answer, that's called fine-tuning a model & it's a very very expensive part of training because you actually require people to create custom data sets & you need to be very careful & curate these datasets very well.

Ex.

① Q/A
What is the capital of Nigeria?
Abuja

② Q/A
What is the capital of Colombia??
Bogotá

③ Q/A
Who discovered algebra?
Al-Khwarizmi

④ Q/A
Who discovered abstract algebra?
Emmy Noether

⇒ This is not only for Question-Answers. If you want the model to actually chat.

Ex. Hello, how are you?
Good, and you?
- - -

Then you need
with tons & to
model starts
it has to
back at you.
of Conversa

Ex. ① Chat
Hello, how are yo
I'm good & yo
Great, thank y

⇒ Another thin
for comman
like Do
OK

① Chat
Write a poem about
elephants
Oh mighty elepha
thou shalt - -

Then you need to fineture the model by post training with tons & tons & tons of chats, In this way model starts learning if you are talking to it & it has to reply as a person who's chatting back at you. & you should remember prev. steps of conversation.

Ex:

① Chat	② Chat	③ Chat	④ Chat
Hello, how are yo?	Good morning, how can I help you?	Hi mom!	Hello please connect me with customer support
I'm good & you?	Thank you, can you connect me with..	Hello dea	of course
Great, thank you			Thank you

⇒ Another thing you should fine tuned the model is for commands.

~~like~~ Do this!

OK, boss!

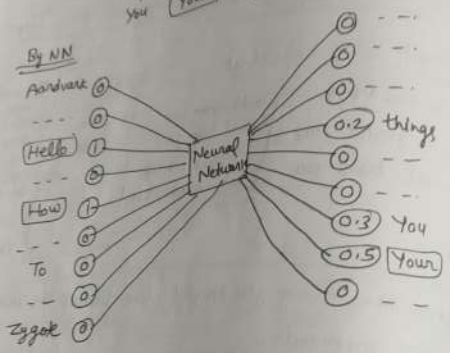
① Chat	② Chat	③ Chat	④ Chat
Write a poem about elephants	Correct this code! print (hellowrld)	Write an essay about middle ages.	Give me a list of fruits.
Oh mighty elephant, thou shalt..	Yes!. You need to add quotations! print ("hello world")	OK! Back in the day	Definitely. Apple Banana Orange.

⇒ How to generate text?

Ex. Hello, how are ___

You / Your / things ?

By NN



↳ But Transformer different from NN, it carries context & also ~~read~~ writes a story by make loop of next word in transformer.

↳ So, Diff. b/w NN & Transformer :-

Transformers have a huge (lots of) parameters & also they are trained on lots of data. & carries context.

NN have feedforward layers ~~and~~ but transformer have diff architecture

(*) "I'm not ... I'm sad" do Positiona...

(*) For multi... embeddings

(*) Post-Traini...

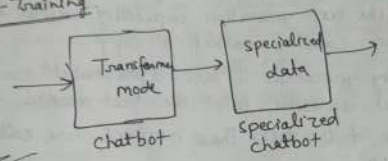
(*) High dot p...

(*) Word & Sente... It is done by ...

(*) "I'm not sad, I'm happy" and "I'm not happy, I'm sad" have same embedding vectors, So, we do Positional encodings.

(*) For multiheaded attention, we use several different embeddings.

(*) Post-Training.



(*) High dot product value means High similarity.

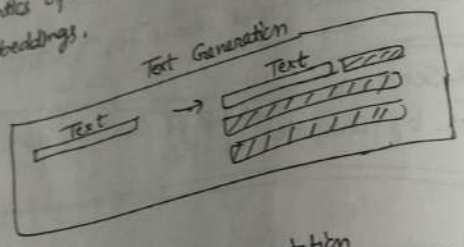
(*) Word & Sentence embeddings can't be done by Human. It is done by Computer by NN.

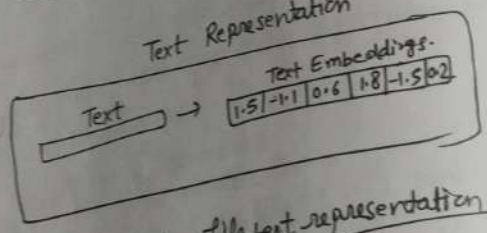
Module ②

Text Representation

When you hear about LLMs, probably the 1st thing that comes to mind is the text generation capability such as writing an essay or creating marketing copy.

Another thing you can get is text representat^n: a set of numbers that represent what the text means & capture the semantics of the text. These numbers are called the text embeddings.





↳ Text generation outputs text, while text representation outputs embeddings.

↳ Text embeddings give you the ability to turn unstructured text data into a structured form. With embeddings, you can compare 2 or more pieces of text, be it single words, sentences, paragraphs or even longer documents.

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↳ Check the Note
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dataset.

⇒ Step① Prep
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–

⇒ Step② Tu

Next, we e
with co.em
embeddings
– texts
– mod

Module ②

Text Representation

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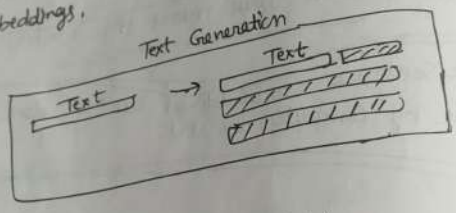

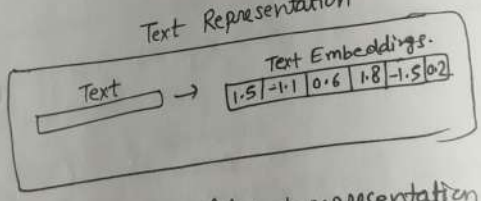

↳ Text generation outputs text, while text representation outputs embeddings.

↳ Text embeddings give you the ability to turn unstructured text data into a structured form. With embeddings, you can compare 2 or more pieces of text, be it single words, sentences, paragraphs or even longer documents.

And since these are sets of numbers, the ways you can process & extract insights from them are limited only by your Imagination.

What does this bring? It opens up many possible use cases that apply in the real world today. Embeddings power app^s we interact with on a daily basis, such as modern search engines, eCommerce product recommendations, social media content moderation, email spam filtering, customer support ~~conversa~~ conversational agents & many more.

↳ Check the Notebook .ipynb file. We'll work at a subset of the Airline Travel Info^n System (ATIS) intent classification dataset.

⇒ Step① Prepare the Dataset

o/p. - which airlines fly from boston to washington dc
- show me the airlines that fly b/w toronto & denver.
- - - .

⇒ Step② Turn text into Embeddings

Next, we embed each inquiry by calling Cohera's Embed endpoint with co.embed(). It takes in texts as input & returns embeddings as output. we supply 3 parameters:

- texts: The list of texts you want to embed
- model: The model to use to generate the embedding.

Till now, there are 4 models available:

- embed-english-v3.0 (English)
- embed-english-light-v3.0 (English)
- embed-multilingual-v3.0 (Multilingual; 100+ languages)
- embed-multilingual-light (v3.0) ...

- input_type - specifies the type of document to be embedded. At the time of writing, there are 4 options:
  - search-document : For documents against which search is performed
  - search-query : For query documents
  - classification : For when the embeddings will be used as an input to a text classifier.
  - Clustering : For when you want to cluster the embeddings.

→ For every piece of text passed to the Embed endpoint, a sequence of 1024 numbers will be generated. Each number represents a piece of info about the meaning contained in that piece of text.

⇒ Step ③ Visualize Embeddings with a Heatmap.

For some visual intuition, compress the dimension to a much lower no., say 10.

Say, we have taken 3 ex
ground transportation i
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⇒ Step ④ : Visuali

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